James Compagno

Exam 2

Please include the relevant R code and output with your answers. Convert the file to a PDF and name it using your first and last name. Upload the PDF to the clearly marked subfolder under Assignments in Canvas by the specified deadline. The quiz is worth 10 points. Good luck!

The internet is NOT allowed for any purpose other than to download and upload the exam.

Answers are in code or blue

1. (5 points) A manager is interested in evaluating the impact of employee training programs on performance. For the dependent variable, they use the lead conversion rate (Rate), defined as the percentage of leads that turn into actual sales. Traditionally, the explanatory variables for the analysis include the experience (Experience, in years) and the training participation (Yes/No) of the employees. (Data: ***Conversion***)
   a. The manager believes that training participation will be particularly effective for employees with less experience and incorporates this belief into the specification of the regression model. Estimate this model to predict the lead conversion rate for an employee with 5 years of experience who has participated in training. What is the corresponding rate with 20 years of experience? Show the R script, the regression output, and the answers.

```
> myData <- read_excel("/Users/jamescompagno/Library/CloudStorage/OneDrive-Personal/MSBA/Econometrics/EconometricsRCode-Data/Data Exam 2.xlsx", sheet = "Conversion")
> myData$Participation <- ifelse(myData$Participation=="Yes",1,0)
> Model1 <- lm(Rate~Experience+Participation,data=myData)
> predict(Model1, data.frame(Experience=5, Participation=1))
1
11.81128
> predict(Model1, data.frame(Experience=20, Participation=1))
1
16.98611
```

   b. Are the predictions in part a consistent with the manager's belief? Provide a brief explanation.

      Yes because while the conversion rate for a 20 year employee with training (16.98) is higher than that a 5 year employee (11.81) it is clear from the answers it does not scale linearly with years of experience.

2. (5 points) Consider the information on 10,000 sales representatives from the hardware and software product groups of a fictional high-tech company. Create a filtered data to include only the <u>software</u> industry, using observations on sales representatives who have a <u>college degree</u> and are <u>older than 40 years</u>. (Data = **Reps**)

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a. How many observations are in the filtered data?
   2187
b. Use the filtered dataset to estimate the regression model: $\ln(\text{Salary}) = \beta_0 + \beta_1\text{Certificates} + \beta_2\text{NPS} + \beta_3\text{Female} + \varepsilon$. Use the estimates to predict the salary for a female representative with 3 certificates and an NPS score of 8. Report the estimates and the prediction.

```
> myData <- read_excel("Data Exam 2.xlsx", sheet = "Reps_Filtered")
> Model1 <- lm(log(Salary) ~ Certificates + NPS + Female, data = myData)
> summary(Model1)

Call:
lm(formula = log(Salary) ~ Certificates + NPS + Female, data = myData)

Residuals:
Min 1Q Median 3Q Max
-0.67169 -0.14438 0.00563 0.14591 0.81770

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 10.795921 0.015015 719.00 <0.0000000000000002 ***
Certificates 0.053083 0.003173 16.73 <0.0000000000000002 ***
NPS 0.057159 0.002436 23.47 <0.0000000000000002 ***
Female -0.130206 0.009411 -13.84 <0.0000000000000002 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.217 on 2183 degrees of freedom
Multiple R-squared: 0.4478, Adjusted R-squared: 0.447
F-statistic: 590 on 3 and 2183 DF, p-value: < 0.00000000000000022
> preexpo <- predict(Model1, data.frame(Certificates = 3, NPS = 8, Female = 1))
> exp(preexpo)
1
79398.52
```

3. (5 points) You would like to examine how individuals' annual spending on restaurant meals responds to changes in meal prices and personal income. The goal is to measure the sensitivity of dining-out behavior to price and income fluctuations, offering insights into

consumer responses to economic conditions and pricing policies. To analyze this relationship, you collect data from 50 individuals and estimate the following log-transformed regression model, with standard errors reported in parentheses.

$$\widehat{ln(\text{Spending})} = \underset{(1.09)}{3.18} - \underset{(0.18)}{0.41}ln(\text{Price}) + \underset{(0.13)}{0.33}ln(\text{Income})$$

a. Interpret the estimated price and income elasticities of spending on restaurant meals.

Price: spending decreases by .41% when price increases by 1%

Income: spending increases by .33% when income increases by 1%

b. At the 5% significance level, test whether the price elasticity of spending differs significantly from zero. Report the appropriate hypotheses, test statistics, p-value, and the inference.

```
# H0: βln(price) = 0

# HA: βln(price) ≠ 0

50 - 2 -1

tb = (-0.41 - 0)/0.18

2*pt(tb, 47, lower.tail=FALSE)

[1] 1.972669

b <- 0.41

se <-

n <- 50; k=length(3)

t <- (b-0)/se; t

pt(t, (n-k), lower.tail = FALSE)

[1] 0.4967454
```

At p-value of 1.97 or 49.67% the elasticity of ln(price) is not significant.

4. (7 points) A researcher wants to examine how immigration affects native-born workers' wages. In theory, a higher labor supply from immigration should lower equilibrium wages in a competitive market. The researcher has data for similarly skilled workers with varying immigrant concentrations. Variables included in the data are Citynum (city of residence), Yrseduc (years of education), Yrsexp (work experience), Ccimmpct (current immigrant share), Cavgtemp (average annual city temperature), Pcimmpct (immigrant share five years earlier, and Wage (hourly wage). (Data = ***Wages***)

a. Estimate the following OLS model:

$$\text{Wage} = \beta_0 + \beta_1\text{Yrseduc} + \beta_2\text{Yrsexp} + \beta_3\text{Ccimmpct} + \varepsilon.$$

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Report the regression results. Is the estimated coefficient on Ccimmpct consistent with the researcher's theory? Explain.

```
> myData <- read_excel("Data Exam 2.xlsx", sheet = "Wages")
> Model_OLS <- lm(Wage~Yrseduc+Yrsexp+Ccimmpct,data=myData)
> summary(Model_OLS)

Call:
lm(formula = Wage ~ Yrseduc + Yrsexp + Ccimmpct, data = myData)

Residuals:
Min 1Q Median 3Q Max
-16.5681 -3.3062 0.0094 3.5128 17.7517

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 11.25017 0.73776 15.25 <0.0000000000000002 ***
Yrseduc 1.28205 0.03996 32.08 <0.0000000000000002 ***
Yrsexp 0.74621 0.01347 55.39 <0.0000000000000002 ***
Ccimmpct -0.19595 0.01140 -17.19 <0.0000000000000002 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.011 on 996 degrees of freedom
Multiple R-squared: 0.8131, Adjusted R-squared: 0.8125
F-statistic: 1444 on 3 and 996 DF, p-value: < 0.00000000000000022
```

Yes the coefficient of Ccimmpct show that an increase in a unit of Ccimmpct decreases hourly wages by -0.196 units

b. The researcher is concerned that the above estimate may be biased if immigrants are not randomly distributed across cities; that is, they are drawn to cities with stronger economies and higher wages. In that case, endogeneity of Ccimmpct could distort the true effect of immigration on wages. Consider an IV estimation with Pcimmpct and Cavgtemp used as instruments for Ccimmpct. Perform a diagnostic test to first ensure the instruments are relevant in the first stage regression of the IV estimation. What does the test imply? Report the appropriate hypotheses, test statistics, *p*-value, and the inference.

Cavgtemp (average annual city temperature)

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```
> Stage1 <- lm(Ccimmpct~Yrseduc+Yrsexp+Cavgtemp,data=myData)
> summary(Stage1)

Call:
lm(formula = Ccimmpct ~ Yrseduc + Yrsexp + Cavgtemp, data = myData)

Residuals:
Min 1Q Median 3Q Max
-33.480 -7.262 -0.756 10.830 22.628

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 81.38524 8.36698 9.727 < 0.0000000000000002 ***
Yrseduc 0.24837 0.10923 2.274 0.0232 *
Yrsexp 0.01681 0.03691 0.456 0.6488
Cavgtemp -0.67645 0.12593 -5.372 0.0000000971 ***
---
Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 13.73 on 996 degrees of freedom
Multiple R-squared: 0.03402, Adjusted R-squared: 0.03111
F-statistic: 11.69 on 3 and 996 DF, p-value: 0.0000001557
> Stage2 <- lm(Wage~Yrseduc+Yrsexp+fitted(Stage1), data=myData)
> summary(Stage2)

Call:
lm(formula = Wage ~ Yrseduc + Yrsexp + fitted(Stage1), data = myData)

Residuals:
Min 1Q Median 3Q Max
-16.7611 -3.8100 0.0733 4.0780 17.1455

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 9.92779 2.96154 3.352 0.000832 ***
Yrseduc 1.27260 0.04974 25.587 < 0.0000000000000002 ***
```

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Yrsexp 0.74559 0.01537 48.525 < 0.0000000000000002 ***

fitted(Stage1) -0.16052 0.07720 -2.079 0.037839 *

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.693 on 996 degrees of freedom

Multiple R-squared: 0.7587, Adjusted R-squared: 0.758

F-statistic: 1044 on 3 and 996 DF, p-value: < 0.00000000000000022

```
# H0: βCavgtemp = 0
# HA: βCavgtemp ≠ 0

> Stage1_Rest <- lm(Ccimmpct~Yrseduc+Yrsexp, data=myData)
> anova(Stage1_Rest, Stage1)
```

Analysis of Variance Table

Model 1: Ccimmpct ~ Yrseduc + Yrsexp

Model 2: Ccimmpct ~ Yrseduc + Yrsexp + Cavgtemp

Res.Df RSS Df Sum of Sq F Pr(>F)

1 997 193165

2 996 187726 1 5438.7 28.856 0.00000009708 ***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Cavgtemp has a significant impact on Ccimmpct and is therefore viable to use as a instrument. It is highly significangt

Pcimmpct (immigrant share five years earlier)

```
> Stage1 <- lm(Ccimmpct~Yrseduc+Yrsexp+Pcimmpct ,data=myData)
> summary(Stage1)
```

Call:

lm(formula = Ccimmpct ~ Yrseduc + Yrsexp + Pcimmpct, data = myData)

Residuals:

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Min 1Q Median 3Q Max

-31.1803 -10.2584 0.3993 12.5936 24.8544

Coefficients:

Estimate Std. Error t value Pr(>|t|)

(Intercept) 30.326854 2.021517 15.002 < 0.0000000000000002 ***

Yrseduc 0.279854 0.108838 2.571 0.0103 *

Yrsexp 0.009757 0.036813 0.265 0.7910

Pcimmpct 0.208852 0.035028 5.962 0.00000000345 ***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 13.68 on 996 degrees of freedom

Multiple R-squared: 0.04029, Adjusted R-squared: 0.0374

F-statistic: 13.94 on 3 and 996 DF, p-value: 0.000000006593

> Stage2 <- lm(Wage~Yrseduc+Yrsexp+fitted(Stage1), data=myData)

> summary(Stage2)

Call:

lm(formula = Wage ~ Yrseduc + Yrsexp + fitted(Stage1), data = myData)

Residuals:

Min 1Q Median 3Q Max

-15.6003 -3.7933 -0.0373 4.0438 17.3584

Coefficients:

Estimate Std. Error t value Pr(>|t|)

(Intercept) 13.77431 2.67937 5.141 0.000000329 ***

Yrseduc 1.30010 0.04870 26.693 < 0.0000000000000002 ***

Yrsexp 0.74739 0.01528 48.921 < 0.0000000000000002 ***

fitted(Stage1) -0.26358 0.06943 -3.796 0.000156 ***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.665 on 996 degrees of freedom

Multiple R-squared: 0.7611, Adjusted R-squared: 0.7604

F-statistic: 1058 on 3 and 996 DF, p-value: < 0.00000000000000022

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```
# H0: βPcimmpct = 0
# HA: βPcimmpct ≠ 0

> Stage1_Rest <- lm(Ccimmpct~Yrseduc+Yrsexp, data=myData)
> anova(Stage1_Rest, Stage1)
Analysis of Variance Table

Model 1: Ccimmpct ~ Yrseduc + Yrsexp
Model 2: Ccimmpct ~ Yrseduc + Yrsexp + Pcimmpct
Res.Df RSS Df Sum of Sq F Pr(>F)
1 997 193165
2 996 186508 1 6657 35.55 0.000000003448 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Pcimmpct has a significant impact on Ccimmpct and is therefore viable to use as a instrument. It is highly significant

c. Estimate the model using instrumental variables (IV). How does the estimated coefficient on Ccimmpct differ from the OLS estimate in part (a)? Report the output and make a brief comment.

Work done above

When Cavgtemp is used as an IV it lessens the impact of the effect of what was Ccimmpct (OLS: -0.19595 vs IVPcimmpct: -0.16052).

When Pcimmpct is used as an IV it increases (by making it more negative) the effect of what was Ccimmpct (OLS: -0.19595 vs IVPcimmpct: -0.26358) but at the cost of increasing standard error.

5. (8 points) A logistic regression model to analyze the role of various factors on the promotion decision of a sales rep in a technology form. The response variable, Promote, indicates whether a sales rep receives a thumbs up or thumbs down for promotion. Predictor variables include age, age-square, female, personality type (use sentinel as the reference category), and the number of certificates held. (Data = ***Promotion***)

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a. Estimate the logistic regression model (Model 1) results and use it to predict the promotion probability of a 40-year-old male rep with three certificates for each of the four personality types. Report the regression results and the predictions.

```
> myData <- read_excel("Data Exam 2.xlsx", sheet = "Promotion")
> myData$Diplomat <- ifelse(myData$Personality=="Diplomat",1,0)
> myData$Explorer <- ifelse(myData$Personality=="Explorer",1,0)
> myData$Sentinel <- ifelse(myData$Personality=="Sentinel",1,0)
> myData$Analyst <- ifelse(myData$Personality=="Analyst",1,0)
> myData$Promote2 <- ifelse(myData$Promote=="Thumbs Up",1,0)
>
Logistic_Model <- glm(Promote2 ~ Age + I(Age^2) + Female + Diplomat + Explorer + Analyst + Certificates, data
= myData,  family = binomial)
> summary(Logistic_Model)

Call:
glm(formula = Promote2 ~ Age + I(Age^2) + Female + Diplomat +
Explorer + Analyst + Certificates, family = binomial, data = myData)

Coefficients:
Estimate Std. Error z value Pr(>|z|)
(Intercept) -10.828069 4.642745 -2.332 0.0197 *
Age 0.295058 0.203915 1.447 0.1479
I(Age^2) -0.003288 0.002270 -1.449 0.1474
Female -0.093371 0.590787 -0.158 0.8744
Diplomat 0.917062 0.884575 1.037 0.2999
Explorer 1.815870 0.910810 1.994 0.0462 *
Analyst 0.127934 1.107343 0.116 0.9080
Certificates 0.886906 0.181222 4.894 0.000000988 ***
---
Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 144.873 on 119 degrees of freedom
Residual deviance: 98.245 on 112 degrees of freedom
AIC: 114.24
```

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Number of Fisher Scoring iterations: 5

```
>
predict(Logistic_Model, data.frame(Age=40, Certificates=3, Female=0, Diplomat=1, Explorer=0, Analyst=0))
1
-0.7094426 #Diplomat
> predict(Logistic_Model, data.frame(Age=40, Certificates=3, Female=0, Diplomat=0, Explorer=1, Analyst=0))
1
0.1893661 #Explorer
> predict(Logistic_Model, data.frame(Age=40, Certificates=3, Female=0, Diplomat=0, Explorer=0, Analyst=0))
1
-1.626504 #Sentinel
> predict(Logistic_Model, data.frame(Age=40, Certificates=3, Female=0, Diplomat=0, Explorer=0, Analyst=1))
1
-1.49857 #Analyst
```

b. What is the optimal age at which a sales rep is most likely to get promoted?

```
> CfLogistic <- coef(Logistic_Model)
> -CfLogistic[2]/(2*CfLogistic[3])
Age
44.86328
```

c. What is the percentage difference in the odds of promotion between an analyst and a sentinel? Report the value and interpret it.

```
>
predA <- predict(Logistic_Model, data.frame(Age=40, Certificates=3, Female=0, Diplomat=0, Explorer=0, Analyst
=1))
>
predS <- predict(Logistic_Model, data.frame(Age=40, Certificates=3, Female=0, Diplomat=0, Explorer=0, Analyst
=0))
> predA - predS
1
0.1279343
```

Analysts have a 12.79% greater chance of getting promoted than a sentinel

d. Estimate and report a restricted logistic model (Model 2) that assumes the influence on the promotion probability is identical for analysts and sentinels. Test this restriction at the

5% level. Report the hypotheses (in terms of the coefficients of the unrestricted Model 1), *p*-value, and the inference.

H0: BAnalyst = BSentinel
HA: BAnalyst != BSentinel

```
>
Model <- glm(Promote2 ~ Age + I(Age^2) + Female + Diplomat + Sentinel + Analyst + Certificates, data = myData,  fa
mily = binomial)
>
ModelR <- glm(Promote2 ~ Age + I(Age^2) + Female + Diplomat + Analyst + Certificates, data = myData,  family = bi
nomial)
> q <- 2
> n <- nobs(Model)
> K <- length(Model$coefficients)
> e1 <- residuals(ModelR)
> se_r <-  sum(e1^2)
> e2 <- residuals(Model)
> se_u <- sum(e2^2)
> Ftest <- ((se_r - se_u)/q)/(se_u/(n-K))
> Ftest
[1] 2.648236
> 2*pf(Ftest, q, n-K, lower.tail = FALSE)
[1] 0.1504118
```

The effects of Analyst and sentinel are not different at the 5% significance level